Dietary phosphorus restriction in dialysis patients: potential impact of processed meat, poultry, and fish products as protein sources.

BACKGROUND: Dietary intake of phosphorus is derived largely from protein sources and is a critical determinant of phosphorus balance in patients with chronic kidney disease. Information about the phosphorus content of prepared foods generally is unavailable, but it is believed to contribute significantly to the phosphorus burden of patients with chronic kidney disease. DESIGN: Analysis of dietary components. SETTING: We measured the phosphorus content of 44 food products, including 30 refrigerated or frozen precooked meat, poultry, and fish items, generally national brands. OUTCOMES: Measured and reported phosphorus content of foods. MEASUREMENTS: Phosphorus by using Association of Analytical Communities official method 984.27; protein by using Association of Analytical Communities official method 990.03. RESULTS: We found that the ratio of phosphorus to protein content in these items ranged from 6.1 to 21.5 mg of phosphorus per 1 g of protein. The mean ratio in the 19 food products with a label listing phosphorus as an additive was 14.6 mg/g compared with 9.0 mg/g in the 11 items without listed phosphorus. The phosphorus content of only 1 precooked food product was available in a widely used dietary database. LIMITATIONS: Results cannot be extrapolated to other products. Manufacturers also may alter the phosphorus content of foods at any time. Protein content was not directly measured for all foods. CONCLUSION: Better reporting of phosphorus content of foods by manufacturers could result in improved dietary phosphorus control without risk of protein malnutrition.